



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMPUTER SCIENCE

PAPER 1 Multiple Choice

4021/1

1 hour

SPECIMEN PAPER

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

INSTRUCTIONS TO CANDIDATES

Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet provided.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark.

Any rough working should be done in this question paper.

This specimen paper consists of 9 printed pages.

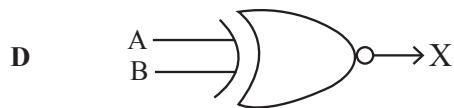
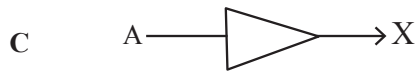
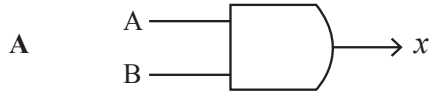
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- 1 A storage technology with fastest access time is
 - A SSD.
 - B HDD.
 - C Optical Disk.
 - D Flash Disk.
- 2 Which software is used to organize and store student data for a school?
 - A Presentation software
 - B Database software
 - C Graphics software
 - D Word processing software
- 3 Utility software is an example of
 - A application software.
 - B customized software.
 - C communication software.
 - D system software.
- 4 HDD stands for
 - A Hardware Disk Drive.
 - B High Definition Device.
 - C Hard Disk Drive.
 - D High Disk Drive.
- 5 Which one is a characteristic of a control system?
 - A Dedicated to one task
 - B Do not involve feedback
 - C Can have several tasks
 - D Lack of the ability to collect and analyse
- 6 Which sensor is used in a greenhouse?
 - A Pressure sensor
 - B Proximity sensor
 - C Acoustic sensor
 - D Humidity sensor
- 7 Android applications are packed in
 - A APK file format.
 - B Axe file format.
 - C ZIP file format.
 - D file format.

8 Flight simulation is used to train pilots because

- A flight simulations are more realistic than flying an actual plane.
- B it enables dangerous flight situations in real life to be modeled.
- C the modeled plane behaves more unpredictably.
- D it is easier to fly a virtual plane.

9 Which of the following logic gates represents a buffer gate?



10 The truth table for an OR gate is

A

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

B

Input A	Input B	Output
0	0	1
0	1	1
1	0	1
1	1	0

C

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	0

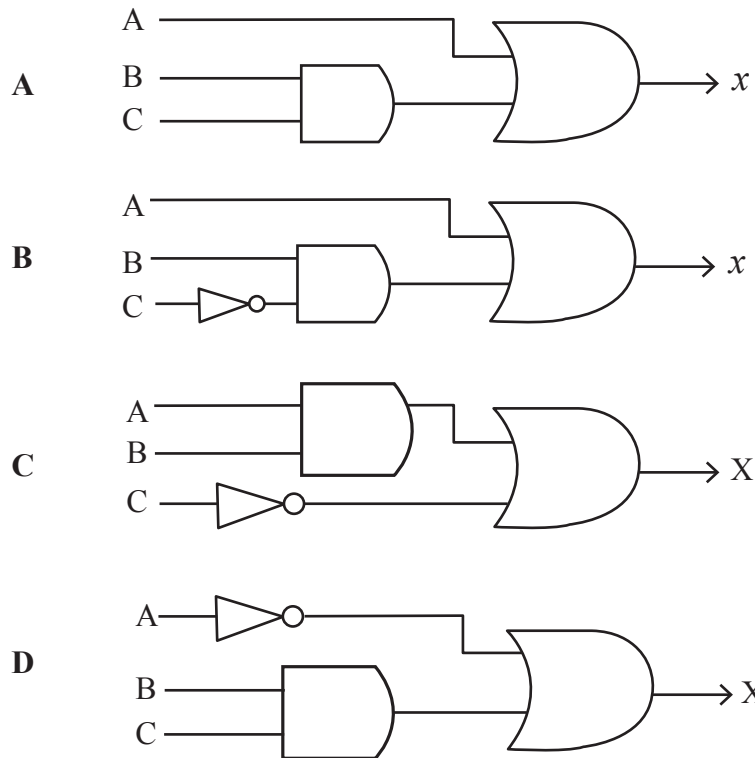
D

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	1

11 A hexadecimal number **AD** in binary is

- A** 10101101
- B** 11011010
- C** 10110101
- D** 10101100

12 Which logic circuit represents the expression $X = A + BC$?



13 A device responsible for finding an optimal route through which to send data packets on a WAN is

- A** modem.
- B** router.
- C** layer-2 switch.
- D** network interface card.

14 A network model suitable for two friends sharing a PDF document using Bluetooth technology is

- A** PAN.
- B** LAN.
- C** Peer to peer.
- D** Client server.

- 15** A computer is unable to access the internet but can communicate with the local network devices.
- What should be checked first?
- A** router configuration
 - B** cable connections
 - C** cookies
 - D** domain name system settings
- 16** Error detection and correction at the destination is done in which layer of the OSI model?
- A** Application
 - B** Network
 - C** Data link
 - D** Physical
- 17** Which security measure is used to ensure data integrity by keying data twice?
- A** Check sum
 - B** Firewall
 - C** Format check
 - D** Verification
- 18** Authentication involves the use of
- A** passwords, biometric sensors and pincodes.
 - B** firewall, magnetic stripes and pincodes.
 - C** biometric sensors, proxy servers and tokens.
 - D** antispyware, passwords and biometric sensors.
- 19** A technique of scrambling data into a format that is not readable by unauthorized users is
- A** updating.
 - B** authorization.
 - C** encryption.
 - D** validation.
- 20** Which expert is hired to test security systems?
- A** white hat hacker
 - B** black hat hacker
 - C** programmer
 - D** cracker

- 21** The testing strategy performed to identify bugs before releasing software to real users or the public is
- A** Top down testing.
 - B** White box testing.
 - C** Alpha testing.
 - D** Beta testing.
- 22** The first step in the system development life cycle is
- A** system design.
 - B** coding.
 - C** system testing.
 - D** preliminary investigation.
- 23** The best test strategy to use when testing a code that deletes an old student record is
- A** integration testing.
 - B** unit testing.
 - C** module testing.
 - D** system testing.
- 24** Which system development stage performs cost benefit analysis?
- A** Design stage
 - B** Feasibility
 - C** Analysis
 - D** Development
- 25** If statements and select case statements are
- A** decision making constraints.
 - B** sequence constraints.
 - C** iteration constraints.
 - D** looping constraints,
- 26** Given the following algorithms
- ```
Start
a = 2
b = 3
c = 4
X = a + b + c
Display X
End
```
- The output is
- A** 12.
  - B** 20.
  - C** 14.
  - D** 11.

27 Given A as 9, A Mod 2 is

- A 4.5.
- B 4.
- C 0.5.
- D 1.

28 Given the following algorithm:

```

Start
Distance = 200 k
Time = 5 hours
Speed = $\frac{\text{distance}}{\text{time}}$

```

Display speed  
End

The speed is

- A 240.
- B 205.
- C 40.
- D 400.

29 Which one is a valid variable name?

- A Employee salary
- B 2<sup>nd</sup>-Number
- C Student Name
- D % Total

30 Which of the following protects the creators from havng their work to be reproduced?

- A copyright
- B patent
- C trademark
- D intellectual property

31 Study the following procedure:

```

Procedure square (X:Integer)
 square = x * x
End procedure

```

The output given the value of x as 3 is

- A 9
- B 0
- C 6
- D 1

- 32 Which database object is used to display information?
- A form
  - B report
  - C query
  - D module
- 33 The validation rule  $\geq 4$  And  $\leq 36$  accepts
- A value 4 and value 9.
  - B any value greater than 4 and less than 36.
  - C either value 4 or value 9
  - D values from 4 to 36
- 34 Which data type accepts alphanumeric data?
- A Memo
  - B Text
  - C Hyperlink
  - D Currency
- 35 The correct query to display all students who were born before 2010 is
- A 1/1/2010.
  - B <1/1/2010.
  - C >1/1/2010.
  - D =1/1/2010.
- 36 An example of software used to create an HTML document is
- A Powerpoint.
  - B Excel.
  - C Notepad.
  - D Publisher.
- 37 Which tags centres the title “Computer Science”?
- A </centre>Computer Science <\centre>
  - B <centre> Conmputer Science <centre>
  - C <centre> Computer Science </centre>
  - D ,centre> Computer Science <centre/>
- 38 The main aim of a business strategy is to
- A review a plan to follow.
  - B improve sales.
  - C gain competitive advantage.
  - D have an overview of the business.



**39**    Start  
      READ amount\_in\_USD  
      READ rate = 32.50 zig  
      Amount\_In\_Zig = amount\_in\_USD \* rate  
      DISPLAY Amount\_In\_Zig  
      Stop

The output for the above pseudocode given that the amount in USD is \$3 is

- A**     97.50.
- B**     35.50.
- C**     97.00.
- D**     96.00.

**40**    The purpose of the pseudocode in question 39 is

- A**     to buy USD.
- B**     to convert currency.
- C**     to withdraw money.
- D**     to find the total amount.